

PARTS LIST

Semiconductors:

- IC1 - L200C adjustable voltage regulator
 IC2 - 7805, 5V voltage regulator
 D1-D2 - 1N5401 rectifier diode
 D3-D6 - 1N4001 rectifier diode
 LED1, LED2 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):

- R1 - 100-ohm, 2-watt
 R2, R5 - 2.2-kilo-ohm
 R3 - 0.47-ohm
 R4 - 1-kilo-ohm
 R6 - 33-ohm, 5-watt
 VR1 - 22-ohm potentiometer
 VR2 - 4.7-kilo-ohm potentiometer
 VR3 - 470-ohm potentiometer

Capacitors:

- C1 - 0.1 μ F, 440V polyester
 C2-C5 - 0.022 μ F ceramic disk
 C6, C8, C9, C12 - 0.33 μ F ceramic disk
 C7 - 2200 μ F, 40V electrolytic
 C10 - 100 μ F, 16V electrolytic
 C11 - 220 μ F, 35V electrolytic

Miscellaneous:

- CON1 - 2-pin connector terminal
 CON2-CON6 - 2-pin connector
 S1 - On/off switch
 F1 - 0.25A fuse
 F2 - 1A fuse
 X1 - 230V AC primary to 18V-0-18V, 1.5A secondary transformer
 - Heat-sinks for L200C and 7805

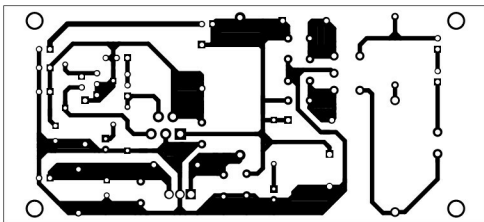


Fig. 2: Actual-size PCB pattern of the dual power supply

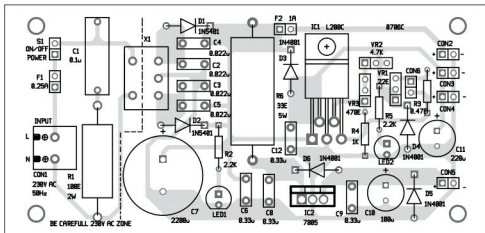


Fig. 3: Component layout of the PCB

pendent of the load until the current limiting is activated. Output at CON5 provides regulated 5V and is protected by IC2 and resistor R6. Wattage of R6 can be increased to 10W if it is heated up excessively.

The output current should not go beyond 300mA. More current is not damaging but the voltage will drop and a lot of power will be dissipated in resistor R6 and IC2. Current-limiting functions of both the regulators are independent.

For testing the circuit, use potentiometer VR1 to control the current. You can measure the voltage drop across R3, which is connected across CON6, using a voltmeter. Using potentiometers VR2 and VR3, vary the voltage at CON3 and CON4. The circuit works on 230V AC, 50Hz as an input connected across CON1. Glowing of LED1 means input is connected. Similarly, glowing of LED2 means output is available for use.

IC1 provides adjustable output from 2.8V up to 18V. IC2 provides independent fixed output of 5V.

Outputs (available at CON3 and CON4) controlled by IC1 are used to charge rechargeable batteries with current from around 10mA up to around 300mA. Maximum voltage can be from around 2.8V up to 18V. Both ICs have inbuilt thermal overload protection.

Construction and testing

An actual-size, single-side PCB for the dual power supply is shown in Fig. 2 and its component layout in Fig. 3.

IC1 and IC2 should be mounted on heat-sinks. Connect the metallic parts (mounting base) of the packages of both the ICs electrically with ground so these can use the same additional heat-sink. In the worst case, power dissipation of IC1 will reach 20W if the current is regulated at 1A and the output is short-circuited.

20W power can still be dissipated by IC1 but not by IC2, which is why resistor R6 is added to help

with power dissipation. Also, output current of IC2 should be limited to around 300mA, which is not a problem in most cases.

It should be taken into consideration that maximum junction temperature of IC1 is $+150^{\circ}\text{C}$. The same parameter for IC2 is $+125^{\circ}\text{C}$. Also, maximum input voltage for IC1 is 40V and for IC2 is 35V. The thermal resistance junction case of the pentawatt package of IC1 is 3°C/W .

Mount switch S1, fuses F1 and F2, indicators LED1 and LED2 on the front side of the panel.

EFY notes. Be careful during the use of circuit as the dotted line on the left shown in the PCB has 230V AC onboard. ●

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